

Vulnerability Assessment Report

Date: 1st September 2026

System Description

The server hardware consists of a high-performance CPU processor and 128 GB of memory. It runs the latest version of the Linux operating system and hosts a MySQL database management system. The server is configured with a stable IPv4 network connection and communicates with other servers across the organization's network. Security measures include TLS-encrypted communications to protect data transmitted between systems.

Scope

The scope of this vulnerability assessment focuses on the current access controls implemented on the database server. The assessment covers a three-month period from June 2026 through August 2026. The NIST SP 800-30 Revision 1 Risk Assessment Guide is used as the framework for identifying threats, assessing risk, and recommending appropriate security controls.

Purpose

The database server stores and manages customer, campaign, and analytical data that supports the organization's business and marketing operations. Because this information is business-critical and contains sensitive data, maintaining its confidentiality, integrity, and availability is essential. This assessment aims to identify potential security risks related to access control and recommend appropriate mitigation strategies.

Risk Assessment

Threat Source	Threat Event	Likelihood / Severity / Risk	Notes
External attacker	Data exfiltration of sensitive information	3 / 3 / 9	High
Insider (Employee)	Disruption of mission-critical operations	2 / 3 / 6	Medium
Unauthorized user	Unauthorized modification or deletion of critical information	1 / 3 / 3	Low

Approach

The assessment evaluated the organization's data storage and management procedures, focusing primarily on existing access controls. Potential threat sources and threat events were identified based on the likelihood of occurrence and the current level of exposure within the information system. The potential business impact of each threat was evaluated

by considering its effect on the confidentiality, integrity, and availability of critical organizational data and services.

Remediation Strategy

- Implement strong authentication mechanisms, including Multi-Factor Authentication (MFA).
- Enforce Role-Based Access Control (RBAC) following the Principle of Least Privilege.
- Enable authentication, authorization, and auditing mechanisms to ensure only authorized users can access the database server.
- Require strong password policies for all user accounts.
- Encrypt all data in transit using TLS 1.2 or TLS 1.3.
- Restrict database access using IP allow-listing and firewall rules.
- Enable centralized logging and continuous monitoring of database activities.
- Regularly apply operating system and database security patches.
- Conduct periodic vulnerability assessments and security audits.

Conclusion

The assessment identified access control as the primary security concern affecting the database server. Although existing security controls provide a basic level of protection, implementing stronger authentication, role-based access control, TLS encryption, centralized logging, and network restrictions will significantly reduce the likelihood and impact of unauthorized access or data compromise. Regular monitoring and periodic security assessments should be performed to maintain a strong security posture.

Skills Demonstrated

Vulnerability Assessment • Risk Analysis • NIST SP 800-30 • Database Security • Access Control • RBAC • Principle of Least Privilege • Threat Modeling • Risk Management • Security Documentation

Key Learning

This activity strengthened my understanding of vulnerability assessments by applying the NIST SP 800-30 framework to identify threats, evaluate organizational risk, and recommend practical security controls. It reinforced the importance of layered security, least-privilege access, and continuous monitoring.